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AnomalyDINO: Boosting Patch-based Few-shot Anomaly Detection with DINOv2

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Abstract

Recent advances in multimodal foundation models have set new standards in few-shot anomaly detection. This paper explores whether high-quality visual features alone are sufficient to rival existing state-of-the-art vision-language models. We affirm this by adapting DINOv2 for one-shot and few-shot anomaly detection, with a focus on industrial applications. We show that this approach does not only rival existing techniques but can even outmatch them in many settings. Our proposed vision-only approach, AnomalyDINO, follows the well-established patch-level deep nearest neighbor paradigm, and enables both image-level anomaly prediction and pixel-level anomaly segmentation. The approach is methodologically simple and training-free and, thus, does not require any additional data for fine-tuning or meta-learning. Despite its simplicity, AnomalyDINO achieves state-of-the-art results in one- and few-shot anomaly detection (e.g., pushing the one-shot performance on MVTec-AD from an AUROC of 93.1% *to* 96.6%*). The reduced overhead, coupled with its outstanding few-shot performance, makes AnomalyDINO a strong candidate for fast deployment, e.g., in industrial contexts.*

1. Introduction

Anomaly detection (AD) in machine learning attempts to identify instances that deviate substantially from the nominal data distribution $p_{\text{norm}}(x)$. Anomalies, therefore, raise suspicion of being 'generated by a different mechanism' [16]—often indicating critical, rare, or unforeseen events. The ability to reliably distinguish anomalies from normal samples is highly valuable across various domains, including security [40], healthcare [14, 41], and industrial inspection. In this work, we focus on the latter, where fully automated systems necessitate the ability to detect defective or missing parts to prevent malfunctions in downstream products, raise alerts for potential hazards, or analyze these to optimize production lines. See the right-hand side of Figure 1 for anomalous samples in this context.

AD for industrial images has gained tremendous interest over the last couple of years. The close-to-optimal results on benchmark data make it seem that the problem of anomaly detection is essentially solved. For instance, Mousakhan et al. [28] report 99.8% and 98.9% AUROC on the popular benchmarks MVTec-AD [2] and VisA [51], respectively. The most popular AD techniques use the training data to train an anomaly classifier [38], or a generative model coupled with reconstruction-based [26, 28, 42], or likelihood-based [9, 35] anomaly scoring. However, these approaches operate within the full-shot setting, meaning they rely on access to a sufficiently large amount of training data. Given the challenges associated with dataset acquisition, the attractivity of a fast and easy-to-deploy methodology, and the requirement to rapidly adapt to covariate shifts in the nominal data distribution [22], there is an increasing interest in few-shot and zero-shot anomaly detection.

Few-shot techniques, however, heavily rely on meaningful features, or as [33] frame it: 'anomaly detection requires better representations'. Such better representations are now available with the increasing availability and capabilities of foundation models, i.e., large-scale models trained on massive datasets in unsupervised/self-supervised fashion [5, 30, 31]. The performance of few-shot anomaly detection techniques has already been boosted by the use of foundation models, mostly by multimodal approaches incorporating language and vision [4, 19, 21, 50].

Here, we propose to focus on a vision-only approach, in contrast to such multimodal techniques. This perspective is motivated by the observation that few-shot anomaly detection is feasible for human annotators based on visual features only, and does not require additional textual descrip-

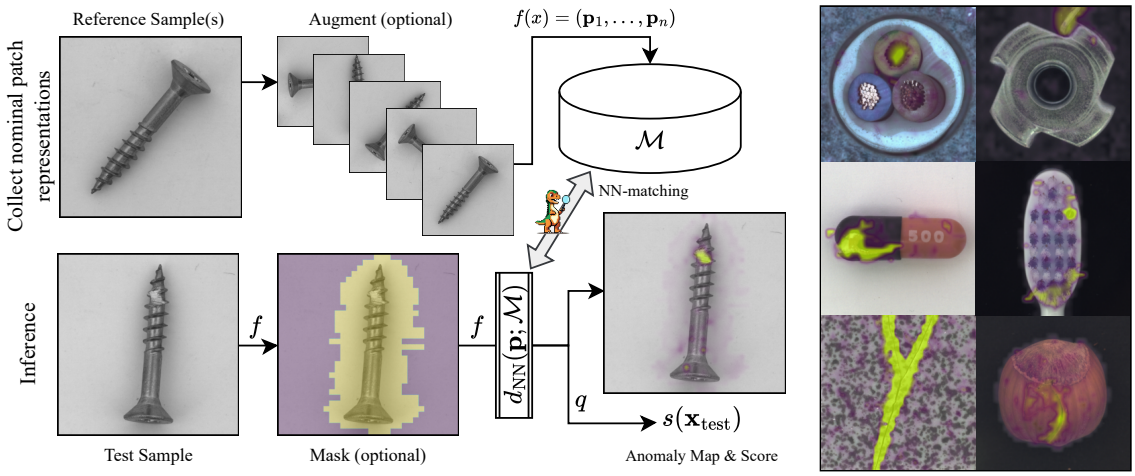


Figure 1. **Anomaly detection with AnomalyDINO** based on a single immaculate reference sample (here category 'Screw' from MVTec-AD). We collect the nominal patch representations from the (potentially augmented) reference sample(s) in the memory bank $\mathcal{M}$. At test time, we select the relevant patch representation via masking (if applicable). The distances of those to the nominal representations in $\mathcal{M}$ give rise to an anomaly map and the corresponding anomaly score $s(\mathbf{x}_{\text{test}})$ using the aggregation statistic q. For both, masking and feature extraction, we utilize DINOv2 (f). Further examples for other categories are depicted on the right (and in Figures 4 to 7 in Appendix A).

tion of the given object or the expected types of anomalies (which are typically not known a priori).

Our approach, termed AnomalyDINO, follows the well-established AD framework of *patch-level deep nearest neighbor* [34, 46], and leverages DINOv2 [30] as a backbone. We carefully design a suitable preprocessing pipeline for the one-shot scenario, using the zero-shot segmentation abilities of DINOv2 (which alleviates the additional overhead of another segmentation model). At test time, anomalous samples are detected based on the high distances between their patch representations and the closest counterparts in the nominal memory bank $\mathcal{M}$.

Due to its simplicity, AnomalyDINO can be deployed in industrial contexts very easily—in strong contrast to more complex approaches such as [7] or [21]. Yet, the proposed method achieves new state-of-the-art performance on anomaly detection in the few-shot regime on MVTec-AD [2] and outperforms all but one competing method on VisA [51].

The structure of the paper is as follows: Section 2 reviews relevant prior studies and clarifies the distinctions between the settings addressed by zero- and few-shot, and batched zero-shot techniques. Section 3 introduces our proposed method, AnomalyDINO. An extension of this method to the batched zero-shot scenario is detailed in Appendix D. Section 4 presents the experimental outcomes. Additional results and an ablation study are provided in Appendices A and C, respectively. We address identified failure cases of AnomalyDINO in Appendix B. The code to reproduce the experiments is available at https://github.com/dammsi/AnomalyDINO.

Contributions

- We propose AnomalyDINO, a simple and training-free yet highly effective patch-based technique for visual anomaly detection. Our method builds upon the high-quality feature representation extracted by DINOv2.
- An extensive analysis demonstrates the efficiency and effectiveness of the proposed approach, outperforming other multimodal few-shot techniques in terms of performance *and* inference speed. Specifically, AnomalyDINO achieves state-of-the-art results for few-shot anomaly detection on MVTec-AD, e.g., pushing the one-shot detection from an AUROC of 93.1% to 96.6% (thereby halving the gap between the few- and full-shot setting). Moreover, our results on VisA are not only competitive with other few-shot methods but also establish a new state-of-the-art for all training-free few-shot anomaly detectors, achieving the best localization performance across all methods.

2. Related Work

Foundation Models for Vision Multimodal foundation models have emerged as powerful tools for a wide range of tasks, see e.g., [3, 6, 17, 23, 25, 29, 31]. Most relevant to visual AD are multimodal approaches based on CLIP [31] or recent LLMs [29], but also vision-only approaches like DINO [5, 30]. CLIP [31] learns visual concepts from natural language descriptions by training on a dataset of images paired with textual annotations. The model uses a contrastive learning objective that aligns the embeddings from

image and text encoders, optimizing the similarity between corresponding image-text pairs. This common feature space for vision and language can be utilized for several downstream tasks, such as zero-shot image classification by assessing similarities to a set of class-specific prompts. DINO [5, 30] leverages a self-supervised student-teacher framework based on vision transformers [12]. It employs a multi-view strategy to train Vision Transformers (ViT) [11] to predict softened teacher outputs, thereby learning robust and high-quality features for downstream tasks. DINOv2 [30] combines ideas from DINO with patch-level reconstruction techniques [49] and scales to larger architectures and datasets. The features extracted by DINO are well-suited for anomaly detection as they incorporate both local and global information, are robust to multiple views and crops, and benefit from large-scale pre-training. GroundingDINO [25], builds upon the DINO framework and focuses on improving the alignment of textual and visual information, enhancing the model's performance in tasks requiring detailed object localization and multimodal understanding.

Anomaly Detection Given a predefined notion of normality, the anomaly detection task is to detect test samples that deviate from this concept [36, 45]. In this work, we focus on *low-level sensory anomalies of industrial image data*, i.e., we do not target the detection of semantic anomalies but of low-level features such as scratches of images of industrial products (see e.g., Figure 1). Several works tackled this task by either training an anomaly classifier [38] or a generative model, which allows for reconstruction-based or likelihood-based AD [9, 26, 28, 42, 48].

A well-established approach is that of *deep nearest neighbor* AD, where test instances are scored according to the distance to their nearest neighbor (in feature space) from the memory bank $\mathcal{M}$, containing features of nominal instances. This approach dates back to (at least) 2002 [13], but got later popularized first on image-level [1] and than by the seminal works on patch-level: Patch support vector data description (SVDD) [46] trains a patch-based feature extractor (with a mixture of DeepSVDD [37] and a self-supervised loss) and PatchCore [34] utilize pre-trained classification models on Imagenet for patch-level feature extraction. Various further approaches build on this simple, but effective AD approach [8, 24, 43]. Besides kNN matching, other 'classical' AD methods like the Mahalanobis distance are employed [10]. Evidently, the performance of these approaches crucially depends on the feature extractor f. Classical choices are ResNets [18], Wide ResNets [47], and Vision Transformers [11], mostly pre-trained on a supervised task.

Another line of work builds upon the success of pre-trained language-vision models in zero-shot classification. The underlying idea consists of two steps. First, these approaches define sets of prompts describing nominal samples and anomalies. Second, the corresponding textual embeddings are compared against the image embeddings [7, 19, 21, 50]. Images whose visual embedding is close to the textual embedding of a prompt associated with an anomaly are classified as anomalous. However, these methods either require significant prompt engineering (e.g., [7] use a total of 35×7 different prompts for describing normal samples) or fine-tuning of the prompt(-embeddings). Lastly, another type of few-shot anomaly detection builds upon the success of multimodal chatbots. These methods require more elaborate prompting and techniques for interpreting textual outputs [44]. Since these methods do not require a memory bank, they are capable of performing zero-shot anomaly detection.

Categorization of Few-/Zero-Shot Anomaly Detectors Previous works consider different AD setups, which complicates their evaluation and comparison. To remedy this, we provide a taxonomy of recent few- and zero-shot AD based on the particular 'shot'-setting, the training requirements, and the modes covered by the underlying models. We categorize three 'shot'-settings: zero-shot, few-shot, and *batched zero-shot*. Zero- and few-shot settings are characterized by the number of nominal training samples a method can process, before making predictions on the test samples. In batched zero-shot, inference is not performed sample-wise but based on a whole batch of test samples, usually the full test set. For instance, the method proposed in [24] benefits from the fact that a significant majority of pixels correspond to normal pixels, which motivates the strategy of matching patches across a batch of images. Another work that considers this setting [22], deploys a parameter-free anomaly detector based on the effect of batch normalization. We split the training requirements into the categories 'Training-Free', 'Fine-Tuning', and 'Meta-

Table 1. **Taxonomy of recent few- and zero-shot anomaly detection methods**. The † indicates approaches that were introduced as full-shot detectors but then considered as few-shot detectors in later works, see e.g., [34].

Method	Setting	Training Type	Modes
DN2 [1]	Few-Shot	Training-Free	Vision
SPADE [8]	Few-Shot†	Training-Free	Vision
PaDiM [10]	Few-Shot†	Fine-Tuning	Vision
PatchCore [34]	Few-Shot	Training-Free	Vision
PatchCore–opt [39]	Few-Shot	Training-Free	Vision
MuSc [24]	Batched Zero-Shot	Training-Free	Vision
GraphCore [43]	Few-Shot	Training (GNN)	Vision
WinCLIP [19]	Zero-/Few-Shot	Training-Free	Vision + Language
APRIL-GAN [7]	Zero-/Few-Shot	Meta-Training	Vision + Language
ADP [21]	Few-Shot	Fine-Tuning	Vision + Language
AnomalyCLIP [50]	Zero-Shot	Meta-Training	Vision + Language
GPT4-V [44]	Zero-Shot	Training-Free	Vision + Language
ACR [22]	Batched Zero-Shot	Meta-Training	/
AnomalyDINO (Ours)	Few-Shot	Training-Free	Vision

Training'. 'Training-Free' approaches do not require any training, while 'Fine-Tuning' methods use the few accessible samples to modify the underlying model. In contrast, 'Meta-Training' is associated with training the model on a dataset related to the test data. For example, [22] train their model on MVTec-AD containing all classes except the class they test against. [50] and [7] train their model on VisA when evaluating the test performance on MVTec-AD and vice versa. Finally, we differentiate the leveraged models, which are either vision models (such as pre-trained ViT) or language-vision models (such as CLIP). We provide a detailed summary in Table 1.

3. Matching Patch Representations for Visual Anomaly Detection

This section introduces AnomalyDINO, which leverages DINOv2 to extract meaningful patch-level features. We build upon the well-established deep nearest neighbor approaches [1, 8, 13, 24, 34, 43, 46], i.e., we first gather relevant patch representations of nominal patches in a memory bank $\mathcal{M}$. Then, for each test patch, we compute its distance to the nearest nominal patch in $\mathcal{M}$. A suitable aggregation of the patch-based distances gives anomaly scores on image-level.

Our work differs from previous deep nearest neighbor approaches [34, 46] by tailoring the memory bank concept to the few-shot regime, specifically utilizing the strong features from DINOv2: We design a pipeline that incorporates *zero-shot masking* and *augmentations*, and we propose a *more robust aggregation statistic*. Importantly, we identify DINOv2 as the ideal backbone for our scenario, due to its strong patch-level features and the masking ability. In addition, this simplifies the deep nearest neighbor framework by reducing the complexity of the feature engineering stage (e.g., by making it unnecessary to think of which representations/layers to use and how to aggregate them), and *increasing the flexibility* w.r.t. input resolution (that can be chosen to be any multiple of 14). The proposed method is described in detail in the following subsections.

3.1. Anomaly Detection via Patch (Dis-) Similarities

Let us briefly review the idea of patch-level deep nearest neighbor AD. We assume to have access to a suitable feature extractor $f : \mathcal{X} \to \mathcal{F}^n$ that maps each image $\mathbf{x} \in \mathcal{X}$ to a tuple of patch-features $f(\mathbf{x}) = (\mathbf{p}_1, \dots, \mathbf{p}_n)$, where $\mathcal{X}$ denotes some space of images and $\mathcal{F}$ the feature space. Note that n depends on the image resolution and the patch size (in the case of DINOv2, a patch is 14×14 pixels). Given $k \geq 1$ nominal reference sample(s) $X_{\text{ref}} := \{\mathbf{x}^{(i)} \mid i \in [k]\}$ (with shorthand $[k] := \{1, \dots, k\}$), we collect the nominal patch features and store them in a memory bank

$$\mathcal{M} := \bigcup_{\mathbf{x}^{(i)} \in X_{\text{ref}}} \left\{ \mathbf{p}_j^{(i)} \mid f(\mathbf{x}^{(i)}) = (\mathbf{p}_1^{(i)}, \dots, \mathbf{p}_n^{(i)}),\, j \in [n] \right\} . \tag{1}$$

To score a test sample $\mathbf{x}_{\text{test}}$, we collect the extracted patch representations $f(\mathbf{x}_{\text{test}}) = (\mathbf{p}_1, \dots, \mathbf{p}_n)$ and then check how well they comply with $\mathcal{M}$. To do so, we leverage a nearest neighbor approach to find the distance to the closest reference patch for a given test patch $\mathbf{p} \in \mathcal{F}$

$$d_{\text{NN}}(\mathbf{p}; \mathcal{M}) := \min_{\mathbf{p}_{\text{ref}} \in \mathcal{M}} d(\mathbf{p},\, \mathbf{p}_{\text{ref}}) \tag{2}$$

for some distance metric d. In our experiments, we set d as the cosine distance, that is,

$$d(\mathbf{x}, \mathbf{y}) := 1 - \frac{\langle \mathbf{x},\, \mathbf{y} \rangle}{\|\mathbf{x}\|\, \|\mathbf{y}\|} \ . \tag{3}$$

The image-level score $s(\mathbf{x}_{\text{test}})$ is given by aggregating the patch distances via a suitable statistic q

$$s(\mathbf{x}_{\text{test}}) := q\big(\{d_{\text{NN}}(\mathbf{p}_1; \mathcal{M}), \dots, d_{\text{NN}}(\mathbf{p}_n; \mathcal{M})\}\big) \ . \tag{4}$$

Throughout this paper, we define q as the average distance of the 1% most anomalous patches, i.e, $q(\mathcal{D}) := \text{mean}(H_{0.01}(\mathcal{D}))$ with $H_{0.01}(\mathcal{D})$ containing the 1% highest values in the set $\mathcal{D}$. The statistic q can be understood as an empirical estimate of the tail value at risk for the 99% quantile [27] and turns out suitable for a wide range of settings because it balances two desirable properties: First, we want $s(\mathbf{x}_{\text{test}})$ to depend on the highest patch distances, as they may provide the strongest anomaly signal. Similarly, we want a certain degree of robustness against a singular high patch distance (in particular in few-shot scenarios where $\mathcal{M}$ is sparsely populated). However, one could also replace q with a different statistic to cater to special cases. If anomalies are expected to cover larger parts of the image, for example, a high percentile of the patch distances might be a suitable choice. If, on the contrary, anomalies may occur only locally such that only very few patches/or a single patch might be affected, the score q should be sensitive to the highest patch distances. Frequently, the maximum pixel-wise anomaly score is considered. Such an anomaly score on pixel-level is usually obtained by upsampling to full image resolution and applying some smoothing operation.

Following [34], we utilize bilinear upsampling and Gaussian smoothing ($\sigma = 4.0$) to turn the patch distances into pixel-level anomaly scores for localization of potential defects. Examples of the resulting anomaly maps are visualized in Figure 1 and Appendix A (Figures 4 to 7).

We extend AnomalyDINO also to the batched zero-shot scenario, see Appendix D (Figure 18).

3.2. Enriching the Memory Bank & Filtering Relevant Patches

In the few-shot anomaly detection setting, the primary challenge is to effectively capture the concept of normality

from the limited set of nominal samples. A useful strategy involves applying simple augmentations, like rotations (following the insights in [43]), to enhance the variability of nominal patch features. This is essential because the variability at test time is likely to be significantly greater than in the reference data, X_{ref}. In contrast, full-shot methods aim to reduce the size of $\mathcal{M}$ to reduce inference time, e.g., [34].

To avoid irrelevant areas of the test image from leading to falsely high anomaly scores, we propose masking the object of interest. This approach mitigates the risk of false positives, particularly in the few-shot regime where the limited reference samples may not adequately capture the natural variations in the background, as exemplified in Figure 10, Appendix C. It is important to note that the appropriate preprocessing technique—whether to apply masking and/or augmentations—should depend on the specific characteristics of the object(s) of interest. For a more detailed discussion on the challenges and considerations in designing an effective preprocessing pipeline, see Appendix C.1.

Masking Masking, i.e., delineating the primary object(s) in an image from its background, can help to reduce false-positive predictions, thereby improving the robustness in the low-data regime. To minimize the overhead of the proposed pipeline, we utilize DINOv2 also for masking. This is achieved by thresholding the first PCA component of the patch features [30]. We observe empirically that this sometimes produces erroneous masks. Frequently, such failure cases occur for close-up shots, where the objects of interest account for $\gtrsim 50\%$ of patches. To address this issue, we check if the PCA-based mask accurately captures the object in the first reference sample and apply the mask accordingly, which gives rise to the 'masking test' in Figure 2. This test is performed *only once* per object as the procedure yields very consistent outputs. In addition, we utilize dilation and morphological closing to eliminate small holes and gaps within the predicted masks. See Figures 14 and 15 for examples of this masking procedure, Table 8 for the outcomes of the masking test per object, and Figure 10 for a visualization of the benefits of masking in the presence of background noise (all in Appendix C). In general, we do not mask textures (e.g., 'Wood' or 'Tile' in MVTec-AD).

Rotation Rotating the reference sample may improve the detection performance by better resembling the variations within the concept of normality captured in $\mathcal{M}$. Consider, e.g., the 'Screw' in MVTec-AD as depicted in Figure 1 for which rotation-invariant features are desirable. On the other hand, rotation can also be detrimental in cases where rotations of (parts of) the object of interest can be considered anomalies themselves (see Figure 13 for an example).

We consider two different settings. In the 'agnostic' case, which we focus on in the main paper, we always augment the reference sample with rotations. We also consider the case where we know about the samples' potential rotations ('informed'). The 'informed' case is sensible as the data collection process can usually be controlled in an industrial/medical setting (or test images can be aligned), and may lead to lower inference time. See Appendix C.1 for the effect of masking and rotation, and a comparison of 'informed' vs. 'agnostic'.



Figure 2. **Masking test on MVTec-AD**. For 'Capsule' and 'Hazelnut' the masking works successfully (top row), while for 'Cable' and 'Transistor' (bottom row) some areas are incorrectly predicted as background that should belong to the object of interest (highlighted in red). See App. C.1 for the outcomes per object.

4. Experiments

AnomalyDINO – Defaults Using DINOv2[1] as backbone allows us to choose from different distillation sizes, which range from small (ViT-S, 21×10^6 parameters) to giant (ViT-G, 1.1×10^9 parameters). To prioritize low latency, we take the smallest model as our default (and denote our default pipeline AnomalyDINO-S, accordingly) and evaluate two input resolutions, 448 and 672 pixels (smaller edge). As discussed above, we utilize the 'agnostic' preprocessing by default (see Appendix C).

Datasets For the experiments, we consider MVTec-AD[2] and VisA,[3] two datasets with high-resolution images for industrial anomaly detection. MVTec-AD consists of fifteen categories, depicting single objects or textures, and up to eight anomaly types per category. VisA has twelve categories and comes with diverse types of anomalies condensed into one anomaly class 'bad'. Some categories in VisA can be considered more challenging as multiple objects and complex structures are present.

Evaluation Metrics We assess the image-level detection and the pixel-level segmentation performance with three metrics each, following [7, 19, 24]. For evaluating the

[1]Code/weights of DINOv2 [30] are available under Apache 2.0 license.
[2]The MVTec-AD dataset [2] is available under CC BY-NC-SA 4.0.
[3]The VisA dataset [51] is released under the CC BY 4.0 license.

detection performance, we measure the area under the receiver-operator curve (AUROC), the F1-score at the optimal threshold (F1-max), and the average precision (AP) using the respective image-level anomaly scores. We quantify the anomaly segmentation performance using the AUROC, F1-max, and the per-region overlap (PRO, [2]) of the segmentation using the pixel-wise anomaly scores. Note that due to a high imbalance of nominal and anomalous pixels—for MVTec-AD we have 97.26% nominal pixel, for VisA even 99.45% [24]—it is not recommended to assess performance solely on segmentation AUROC [32]. We repeat each experiment three times and report mean and standard deviation.[4]

Baselines We compare AnomalyDINO with a range of modern zero- and few-shot AD models, e.g., SPADE [8], PaDiM [10], PatchCore [34], WinCLIP+ [19], and APRIL-GAN [7]. It is important to note that ACR [22] and MuSc [24] consider the batched zero-shot setting (see Table 1), thus, covering a different setting than AnomalyDINO. We adapt AnomalyDINO to this setting in Appendix D. Moreover, APRIL-GAN and AnomalyCLIP, require training on a related dataset, which is in contrast to our proposed training-free method. In Tables 2 and 3, results reported from [19] are indicated by †, the result of the WinCLIP re-implementation [21] (where J. Jeong is also coauthor) by *. All other results are taken from the original publication. For ADP [21] we report the (usually slightly better) variant ADP$_\ell$ (with access to the class label). For GraphCore no standard deviations are reported [43].

4.1. Few-Shot Anomaly Detection and Anomaly Segmentation

Results We summarize the results for few-shot anomaly detection on MVTec-AD and VisA in Table 2 and Table 3, respectively. Regarding MVTec-AD, our method achieves state-of-the-art k-shot detection performance across all $k \in \{1, 2, 4, 8, 16\}$ for every reported metric, outperforming approaches that require additional training data sets (such as ADP and APRIL-GAN). The method also demonstrates superior anomaly localization, with results showing that while detection performance remains comparable across different resolutions (448 vs. 672), a higher resolution enhances localization performance. Furthermore, we observe clear improvements as the number of samples increases.

In terms of anomaly detection in the VisA benchmark (Table 3), APRIL-GAN demonstrates the strongest performance for $k \in \{1, 2\}$. Nonetheless, AnomalyDINO consistently achieves second-best results for $k \in \{1, 2\}$, comparable results in the 4-shot setting, and sets new state-of-the-art for $k \in \{8, 16\}$. This can be attributed to AnomalyDINO's ability to benefit more from a richer memory bank $\mathcal{M}$ than APRIL-GAN. We hypothesize that meta-learning exerts a greater influence on APRIL-GAN (i.e., training on MVTec-AD, when testing on VisA, and vice-versa) compared to learning from the nominal features of the given reference samples. Note also that AnomalyDINO outperforms all other training-free approaches.

Comparing the segmentation performance on VisA, Table 3 reveals a clear picture: AnomalyDINO consistently shows the strongest localization performance in all metrics considered. While AnomalyDINO-S (448) already demonstrates strong performance, the advantages of using a higher resolution (672) become more evident on the VisA dataset. We attribute this fact to smaller anomalous regions (for which smaller effective patch sizes are beneficial) and more complex scenes (compared to MVTec-AD).

Table 2. **Anomaly detection on MVTec-AD**. Quantitative results for detection (image-level) and segmentation (pixel-level). For each shot, we highlight the **best result** in bold, the results from the second best method as underlined, and the best training-free result by a gray box (see also Table 1). All results in %.

Method	Classification			Segmentation		
	AUROC	F1-max	AP	AUROC	F1-max	PRO
1-shot						
SPADE†	82.9$_{\pm 2.6}$	91.1$_{\pm 1.0}$	91.7$_{\pm 1.2}$	92.0$_{\pm 0.3}$	44.5$_{\pm 1.0}$	85.7$_{\pm 0.7}$
PatchCore†	83.4$_{\pm 3.0}$	90.5$_{\pm 1.5}$	92.2$_{\pm 1.5}$	92.0$_{\pm 1.0}$	50.4$_{\pm 2.1}$	79.7$_{\pm 2.0}$
GraphCore	89.9$_{\pm \ /}$	/	/	95.6$_{\pm \ /}$	/	/
WinCLIP+	93.1$_{\pm 2.0}$	93.7$_{\pm 1.1}$	96.5$_{\pm 0.9}$	95.2$_{\pm 0.5}$	55.9$_{\pm 2.7}$	87.1$_{\pm 1.2}$
APRIL-GAN	92.0$_{\pm 0.3}$	92.4$_{\pm 0.2}$	95.8$_{\pm 0.2}$	95.1$_{\pm 0.1}$	54.2$_{\pm 0.0}$	90.6$_{\pm 0.2}$
AnomalyDINO-S (448)	96.5$_{\pm 0.4}$	**96.0**$_{\pm 0.2}$	98.1$_{\pm 0.3}$	96.3$_{\pm 0.1}$	57.9$_{\pm 0.8}$	91.7$_{\pm 0.1}$
AnomalyDINO-S (672)	**96.6**$_{\pm 0.4}$	95.8$_{\pm 0.5}$	**98.2**$_{\pm 0.2}$	**96.8**$_{\pm 0.1}$	**60.2**$_{\pm 1.1}$	**92.7**$_{\pm 0.1}$
2-shot						
SPADE†	81.0$_{\pm 2.0}$	90.3$_{\pm 0.8}$	90.6$_{\pm 0.8}$	91.2$_{\pm 0.4}$	42.4$_{\pm 1.0}$	83.9$_{\pm 0.7}$
PatchCore†	86.3$_{\pm 3.3}$	92.0$_{\pm 1.5}$	93.8$_{\pm 1.7}$	93.3$_{\pm 0.6}$	53.0$_{\pm 1.7}$	82.3$_{\pm 1.3}$
GraphCore	91.9$_{\pm \ /}$	/	/	96.9$_{\pm \ /}$	/	/
WinCLIP+	94.4$_{\pm 1.3}$	94.4$_{\pm 0.8}$	97.0$_{\pm 0.7}$	96.0$_{\pm 0.3}$	58.4$_{\pm 1.7}$	88.4$_{\pm 0.9}$
ADP$_\ell$	95.4$_{\pm 0.9}$	/	/	/	/	/
APRIL-GAN	92.4$_{\pm 0.3}$	92.6$_{\pm 0.1}$	96.0$_{\pm 0.2}$	95.5$_{\pm 0.0}$	55.9$_{\pm 0.5}$	91.3$_{\pm 0.1}$
AnomalyDINO-S (448)	96.7$_{\pm 0.8}$	**96.5**$_{\pm 0.4}$	98.1$_{\pm 0.7}$	96.5$_{\pm 0.2}$	58.5$_{\pm 0.5}$	92.0$_{\pm 0.2}$
AnomalyDINO-S (672)	**96.9**$_{\pm 0.7}$	96.1$_{\pm 0.3}$	**98.2**$_{\pm 0.5}$	**97.0**$_{\pm 0.2}$	**61.0**$_{\pm 0.5}$	**93.1**$_{\pm 0.2}$
4-shot						
SPADE†	84.8$_{\pm 2.5}$	91.5$_{\pm 0.9}$	92.5$_{\pm 1.2}$	92.7$_{\pm 0.3}$	46.2$_{\pm 1.3}$	87.0$_{\pm 0.5}$
PatchCore†	88.8$_{\pm 2.6}$	92.6$_{\pm 1.6}$	94.5$_{\pm 1.5}$	94.3$_{\pm 0.5}$	55.0$_{\pm 1.9}$	84.3$_{\pm 1.6}$
GraphCore	92.9$_{\pm \ /}$	/	/	**97.4**$_{\pm \ /}$	/	/
WinCLIP+	95.2$_{\pm 1.3}$	94.7$_{\pm 0.8}$	97.3$_{\pm 0.6}$	96.2$_{\pm 0.3}$	59.5$_{\pm 1.8}$	89.0$_{\pm 0.8}$
ADP$_\ell$	96.2$_{\pm 0.8}$	/	/	/	/	/
APRIL-GAN	92.8$_{\pm 0.2}$	92.8$_{\pm 0.1}$	96.3$_{\pm 0.1}$	95.9$_{\pm 0.0}$	56.9$_{\pm 0.1}$	91.8$_{\pm 0.1}$
AnomalyDINO-S (448)	97.6$_{\pm 0.1}$	**97.0**$_{\pm 0.3}$	98.4$_{\pm 0.3}$	96.7$_{\pm 0.1}$	59.2$_{\pm 0.4}$	92.4$_{\pm 0.1}$
AnomalyDINO-S (672)	**97.7**$_{\pm 0.2}$	96.6$_{\pm 0.0}$	**98.7**$_{\pm 0.1}$	97.2$_{\pm 0.1}$	**61.8**$_{\pm 0.1}$	**93.4**$_{\pm 0.1}$
8-shot						
GraphCore	95.9$_{\pm \ /}$	/	/	**97.8**$_{\pm \ /}$	/	/
WinCLIP+	94.6$_{\pm 0.1}$*	/	/	/	/	/
ADP$_\ell$	97.0$_{\pm 0.2}$	/	/	/	/	/
APRIL-GAN	93.1$_{\pm 0.2}$	93.1$_{\pm 0.2}$	96.4$_{\pm 0.2}$	96.2$_{\pm 0.1}$	57.7$_{\pm 0.2}$	92.4$_{\pm 0.2}$
AnomalyDINO-S (448)	98.0$_{\pm 0.1}$	**97.4**$_{\pm 0.1}$	99.0$_{\pm 0.2}$	97.0$_{\pm 0.1}$	59.6$_{\pm 0.2}$	92.7$_{\pm 0.0}$
AnomalyDINO-S (672)	**98.2**$_{\pm 0.2}$	**97.4**$_{\pm 0.2}$	**99.1**$_{\pm 0.1}$	97.4$_{\pm 0.1}$	**62.3**$_{\pm 0.1}$	**93.8**$_{\pm 0.1}$
16-shot						
WinCLIP+	94.8$_{\pm 0.1}$*	/	/	/	/	/
ADP$_\ell$	97.0$_{\pm 0.3}$	/	/	/	/	/
APRIL-GAN	93.2$_{\pm 0.1}$	93.0$_{\pm 0.1}$	96.5$_{\pm 0.1}$	96.4$_{\pm 0.0}$	58.5$_{\pm 0.1}$	92.6$_{\pm 0.1}$
AnomalyDINO-S (448)	98.3$_{\pm 0.1}$	**97.7**$_{\pm 0.2}$	99.3$_{\pm 0.0}$	97.1$_{\pm 0.1}$	60.0$_{\pm 0.1}$	92.9$_{\pm 0.1}$
AnomalyDINO-S (672)	**98.4**$_{\pm 0.1}$	97.6$_{\pm 0.1}$	**99.3**$_{\pm 0.0}$	**97.5**$_{\pm 0.0}$	**62.7**$_{\pm 0.1}$	**94.0**$_{\pm 0.1}$

[4] The randomness stems from the choice of reference samples X_{ref}. For straightforward reproducibility, we simply set X_{ref} to the first k, second k and third k train samples for the three different runs, respectively.

Table 3. **Anomaly detection on VisA**. Quantitative results for detection (image-level) and segmentation (pixel-level). For each shot, we highlight the **best result** in bold, the results from the second best method as underlined, and the best training-free result by a gray box (see also Table 1). All results in %.

Method	Classification			Segmentation		
	AUROC	F1-max	AP	AUROC	F1-max	PRO
1-shot						
SPADE†	79.5±4.0	80.7±1.9	82.0±3.3	95.6±0.4	35.5±2.2	84.1±1.6
PaDiM†	62.8±5.4	75.3±1.2	68.3±4.0	89.9±0.8	17.4±1.7	64.3±2.4
PatchCore†	79.9±2.9	81.7±1.6	82.8±2.3	95.4±0.6	38.0±1.9	80.5±2.5
WinCLIP+	83.8±4.0	83.1±1.7	85.1±4.0	96.4±0.4	41.3±2.3	85.1±2.1
APRIL-GAN	**91.2**±0.8	**86.9**±0.6	**93.3**±0.8	96.0±0.0	38.5±0.3	90.0±0.1
AnomalyDINO-S (448)	85.6±1.5	83.1±1.1	86.6±1.3	97.5±0.1	41.9±0.5	90.7±0.5
AnomalyDINO-S (672)	87.4±1.2	84.3±0.5	89.0±1.0	**97.8**±0.1	**45.1**±0.9	**92.5**±0.5
2-shot						
SPADE†	80.7±5.0	81.7±2.5	82.3±4.3	96.2±0.4	40.5±3.7	85.7±1.1
PaDiM†	67.4±5.1	75.7±1.8	71.6±3.8	92.0±0.7	21.1±2.4	70.1±2.6
PatchCore†	81.6±4.0	82.5±1.8	84.8±3.2	96.1±0.5	41.0±3.9	82.6±2.3
WinCLIP+	84.6±2.4	83.0±1.4	85.8±2.7	96.8±0.3	43.5±3.3	86.2±1.4
ADP$_\ell$	86.9±0.9	/	/	/	/	/
APRIL-GAN	**92.2**±0.3	**87.7**±0.3	**94.2**±0.3	96.2±0.0	39.3±0.2	90.1±0.1
AnomalyDINO-S (448)	88.3±1.8	84.8±1.2	89.2±1.3	97.8±0.1	44.2±0.3	91.7±0.5
AnomalyDINO-S (672)	89.7±1.3	86.3±1.2	90.7±0.8	**98.0**±0.1	**47.6**±0.5	**93.4**±0.6
4-shot						
SPADE†	81.7±3.4	82.1±2.1	83.4±2.7	96.6±0.3	43.6±3.6	87.3±0.8
PaDiM†	72.8±2.9	78.0±1.2	75.6±2.2	93.2±0.5	24.6±1.8	72.6±1.9
PatchCore†	85.3±2.1	84.3±1.3	87.5±2.1	96.8±0.3	43.9±3.1	84.9±1.4
WinCLIP+	87.3±1.8	84.2±1.6	88.8±1.8	97.2±0.2	47.0±3.0	87.6±0.9
ADP$_\ell$	88.4±0.4	/	/	/	/	/
APRIL-GAN	**92.6**±0.4	88.4±0.5	**94.5**±0.3	96.2±0.0	40.0±0.1	90.2±0.1
AnomalyDINO-S (448)	91.3±0.8	87.5±1.0	91.8±0.7	98.0±0.0	46.1±0.3	92.5±0.2
AnomalyDINO-S (672)	**92.6**±0.9	**88.8**±0.9	92.9±0.7	**98.2**±0.0	**49.4**±0.3	**94.1**±0.1
8-shot						
WinCLIP+	85.0±0.0*	/	/	/	/	/
ADP$_\ell$	89.2±0.1	/	/	/	/	/
APRIL-GAN	93.0±0.2	88.8±0.2	**94.9**±0.3	96.3±0.0	40.2±0.1	90.2±0.0
AnomalyDINO-S (448)	92.6±0.1	88.6±0.2	92.9±0.2	98.2±0.0	47.6±0.5	93.3±0.2
AnomalyDINO-S (672)	**93.8**±0.3	**90.0**±0.1	94.3±0.4	**98.4**±0.0	**51.1**±0.4	**94.8**±0.2
16-shot						
WinCLIP+	85.0±0.1*	/	/	/	/	/
ADP$_\ell$	90.1±0.5	/	/	/	/	/
APRIL-GAN	93.2±0.2	89.0±0.1	95.2±0.1	96.3±0.0	40.6±0.1	90.2±0.1
AnomalyDINO-S (448)	93.8±0.1	89.9±0.3	94.2±0.3	98.3±0.0	48.6±0.3	93.8±0.2
AnomalyDINO-S (672)	**94.8**±0.2	**90.9**±0.2	**95.3**±0.3	**98.5**±0.0	**52.5**±0.5	**95.3**±0.2

We have further adapted AnomalyDINO to the batched zero-shot setting (see Appendix D). This adaptation is straightforward and relatively simple, especially compared to MuSc. Notably, we did not employ additional techniques such as 'Re-scoring with Constrained Image-level Neighborhood' or 'Local Neighborhood Aggregation with Multiple Degrees' [24]. The results obtained, as detailed in Table 4, are therefore quite satisfactory.

We conclude that AnomalyDINO—despite its simplicity—rivals other methods in all settings and even comes out ahead in most of the settings (e.g., significantly reducing the gap between few-shot and full-shot methods for MVTec-AD). Within the class of training-free models, it is the clear winner essentially across the board. These results do not only demonstrate the virtues of AnomalyDINO itself but, more generally, highlight the merits of strong visual features as compared to highly engineered architectures. We provide further qualitative results in Appendix A, and discuss the limitations (such as detecting semantic anomalies) and specific failure cases of our approach in Appendix B.

Table 4. **Detection results for other settings**. All results are AUROC values (in %).

Setting	Method	MVTec-AD	VisA
0-shot	WinCLIP	91.8	78.1
	AnomalyCLIP	91.5	82.1
	APRIL-GAN	86.1	78.0
Batched 0-shot	ACR	85.8	/
	MuSc	97.8	92.8
	AnomalyDINO-S (448)	93.0	89.7
	AnomalyDINO-S (672)	94.2	90.7

4.2. Ablation Study

We conduct additional experiments to assess the effect of specific design choices in our pipeline. The full ablation study is provided in Appendix C, here we briefly summarize the main insights. Figure 3 supports the ablation study by highlighting the two key aspects: performance and runtime.

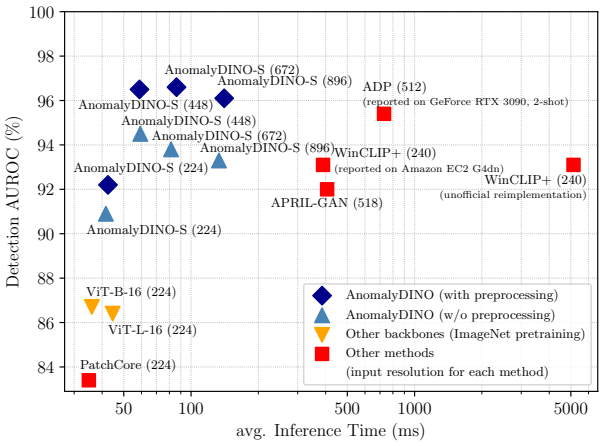


Figure 3. **Detection AUROC vs. inference time** per sample on MVTec-AD in the 1-shot setting. The input resolution is given in parentheses after the method name. All runtimes are measured on a single NVIDIA A40 if not stated otherwise. (Note that for ADP and WinCLIP+ no official code is available.)

Inference time Comparing the inference times of AnomalyDINO with the other approaches in the 1-shot setting (see Figure 3), we observe that AnomalyDINO achieves significantly faster inference times than SOTA few-shot competitors (note the logarithmic scale). For example, AnomalyDINO-S takes roughly 60ms to process an image at a resolution of 448. The only approaches with lower inference time are PatchCore and AnomalyDINO with ViT-backbones trained on ImageNet, which however

sacrifice performance. Notably, while augmentations increase the memory bank, and thus, the potential runtime of the nearest neighborhood search, we find that the effect is negligible in practice.[5] A more detailed runtime analysis is included in Appendix C, Table 9.

Preprocessing To study the impact of preprocessing, we compare the resulting detection AUROCs of AnomalyDINO with and without preprocessing in Figure 3. Across four different configurations of AnomalyDINO, we observe that by incorporating the proposed preprocessing, the resulting AUROCs increase by approximately 2% without significantly affecting the inference time. The effect of proper preprocessing seems to increase with higher resolutions. Additionally, we compare the two different preprocessing settings, 'agnostic' (our default) and 'informed', and find that the agnostic preprocessing slightly outperforms the informed counterpart, see Figures 11 and 12. Depending on the product category, suitable preprocessing steps lead to substantial improvements. And while in principle augmenting the reference sample with rotations may negatively impact the detection performance, we observe this in only very few examples. For the full discussion, see Appendix C.1.

Aggregation statistic We compare the default scoring method, the empirical tail value at risk, to other suitable aggregations in Appendix C.2. The mean of the 1% highest distances from test patches to $\mathcal{M}$ improves over the standard choice (maximum of the upsampled and smoothed patch distances).

Architecture size/choice We also evaluated the proposed framework to Vision Transformers pre-trained on ImageNet, see Figure 3 and Appendix C.4. While these backbones slightly outperform PatchCore, they give weaker features compared to DINOv2 and are incompatible with the proposed masking procedure. This underlines that DINOv2 is indeed excellently suited for visual AD. In addition, ViT-based architectures like DINOv2 easily allow handling images with varying resolutions. In this context, we find that operating at a resolution of 448 gives the best trade-off between performance and inference time, but even higher detection AUROCs are possible at higher resolutions. We also evaluated our pipeline with DINOv2 at different distillation sizes (ViT-S, ViT-B, ViT-L). The full results are given in Appendix C.3, Figure 16. We observe no considerable differences, in particular, also larger backbones give state-of-the-art results on MVTec-AD (all k) and VisA ($k \geq 8$). Interestingly, larger architectures do not necessarily translate to higher performance. The smallest model demonstrates the best performance on MVTec-AD, which primarily consists of single objects and less complex scenes. In contrast, the larger architecture sizes perform better on VisA, which often involves multiple and more complex objects. This suggests the importance of achieving the 'just right' level of abstraction for the specific task, particularly in the few-shot regime.

[5] Our implementation leverages GPU-accelerated neighborhood search [20], leading to only slightly increased inference time. With increasing size of $\mathcal{M}$ reduction techniques such as coreset subsampling [34] are advisable.

5. Conclusion

This paper proposes a vision-only approach for one- and few-shot anomaly detection. Our method is based on similarities between patch representations extracted by DINOv2 [30]. We carefully design means to populate the nominal memory bank with diverse and relevant features while minimizing false positives utilizing zero-shot segmentation and simple data augmentation. Industrial settings require fast deployment, easy debugging and error correction, and rapid adaptation for covariance shifts in the normal data distribution. Our pipeline caters to these requirements through its simplicity and computational efficiency. The proposed method, AnomalyDINO, achieves state-of-the-art results on MVTec-AD and rivals all competing methods on VisA while outperforming all other training-free approaches. Thus, our approach achieves the best of both worlds: improved performance *and* reduced inference time, especially compared to more complex vision-language methods. Its simplicity and strong performance render AnomalyDINO an excellent candidate for practitioners for industrial anomaly detection and an effective baseline for assessing few-shot and even full-shot anomaly detection in ongoing research.

Follow-up research directions The specific pipeline exemplified in the paper focuses on simplicity and high throughput. However, the individual parts of our pipeline can easily be exchanged for more sophisticated alternatives. For example, our simple masking approach could be replaced by more specialized and adaptive masking techniques (which may also be relevant to other methods building on DINOv2), and the simple upsampling and smoothing approach could be substituted by more sophisticated methods like [15]. It would be interesting to see if that leads to further improvements in anomaly detection and localization, thereby further reducing the gap between few- and full-shot anomaly detectors. We also plan to improve the batched zero-shot performance of AnomalyDINO.

Acknowledgement The authors acknowledge funding from TRR 391 *Spatio-temporal Statistics for the Transition of Energy and Transport* by the German Research Foundation (DFG).

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